Build the Express.js backend

Create a new folder called `**server**` under your **agentai** folder (NOTE: it is **NOT** under /src).

We will build our Express.js backend in this `**server**` folder, after it's done, it will look something like this. No rush to create this structure now. We will do it step by step in this class.

```
server/
├── chat.js
├── package-lock.json
├── package.json
├── .env
├── server.js
└── uploads
    ├── hbs-lean-startup.pdf
    └── any-pdf-file-of-your-choice.pdf
```

First, let's create server/package.json file with the following content

```
{
  "name": "server",
  "version": "1.0.0",
  "description": "",
  "main": "server.js",
  "scripts": {
    "start": "node server.js",
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "@langchain/classic": "^1.0.5",
    "@langchain/community": "^1.0.5",
    "@langchain/core": "^1.1.0",
    "@langchain/openai": "^1.1.3",
    "@modelcontextprotocol/sdk": "^1.23.0",
    "cors": "^2.8.5",
    "dotenv": "^16.3.1",
    "express": "^4.18.2",
    "langchain": "^1.1.1",
    "multer": "^1.4.5-lts.1",
    "pdf-parse": "^1.1.1",
    "serpapi": "^2.2.1",
    "zod": "^4.1.13"
  },
  "type": "module"
}
```

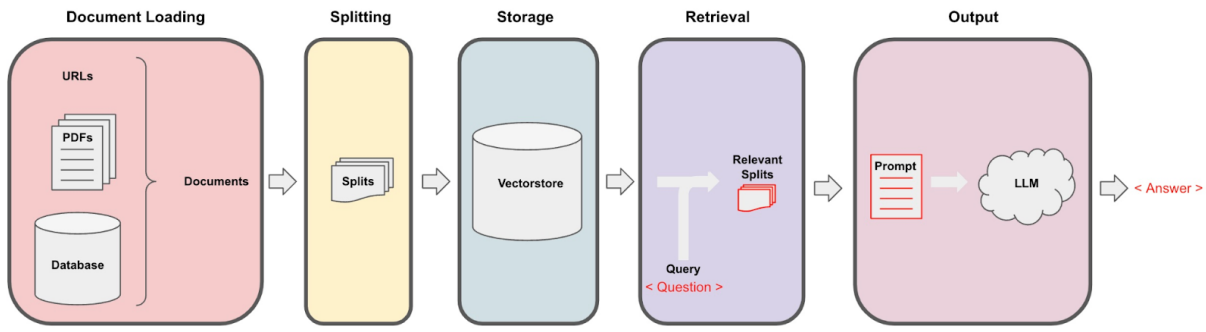
- Run `**npm install**`, under “/server” path
- Create a folder called `**uploads**` inside the /server**,** it will save the uploaded files to this folder
- Create a file called **.env** with the following content
  OPENAI_API_KEY=<YOUR_API_KEY>

  e.g.
  OPENAI_API_KEY=sk-ABCDEFG

Next, find the .gitignore file in the root and add server/.env to it

This is the sample file if you ever need it [hbs-lean-startup.pdf]

Create a file called **chat.js** with the following content




```
import { RecursiveCharacterTextSplitter } from "@langchain/textsplitters";
import { OpenAIEmbeddings } from "@langchain/openai";
import { MemoryVectorStore } from "@langchain/classic/vectorstores/memory";
import { ChatOpenAI } from "@langchain/openai";
import { PromptTemplate } from "@langchain/core/prompts";
import { PDFLoader } from "@langchain/community/document_loaders/fs/pdf";

// NOTE: change this default filePath to any of your default file name
const chat = async (filePath = "./uploads/hbs-lean-startup.pdf", query) => {
  // Get API key from environment
  const apiKey = process.env.OPENAI_API_KEY;

  // step 1:
  const loader = new PDFLoader(filePath);

  const data = await loader.load();

  // step 2:
  const textSplitter = new RecursiveCharacterTextSplitter({
    chunkSize: 500, //  (in terms of number of characters)
    chunkOverlap: 0,
  });

  const splitDocs = await textSplitter.splitDocuments(data);

  // step 3

  const embeddings = new OpenAIEmbeddings(apiKey ? { apiKey } : {});

  const vectorStore = await MemoryVectorStore.fromDocuments(
    splitDocs,
    embeddings
  );

  // step 4: retrieval

  // const relevantDocs = await vectorStore.similaritySearch(
  // "What is task decomposition?"
  // );

  // step 5: qa w/ customize the prompt
  const model = new ChatOpenAI({
    model: "gpt-5",
    ...(apiKey && { apiKey }),
  });

  const template = `Use the following pieces of context to answer the question at the end.
If you don't know the answer, just say that you don't know, don't try to make up an answer.
Use three sentences maximum and keep the answer as concise as possible.

{context}
Question: {question}
Helpful Answer:`;

  const prompt = PromptTemplate.fromTemplate(template);

  // Use retriever to get relevant documents
  const retriever = vectorStore.asRetriever();
  const relevantDocs = await retriever.invoke(query);

  // Format context from retrieved documents
  const context = relevantDocs.map((doc) => doc.pageContent).join("\n\n");
```

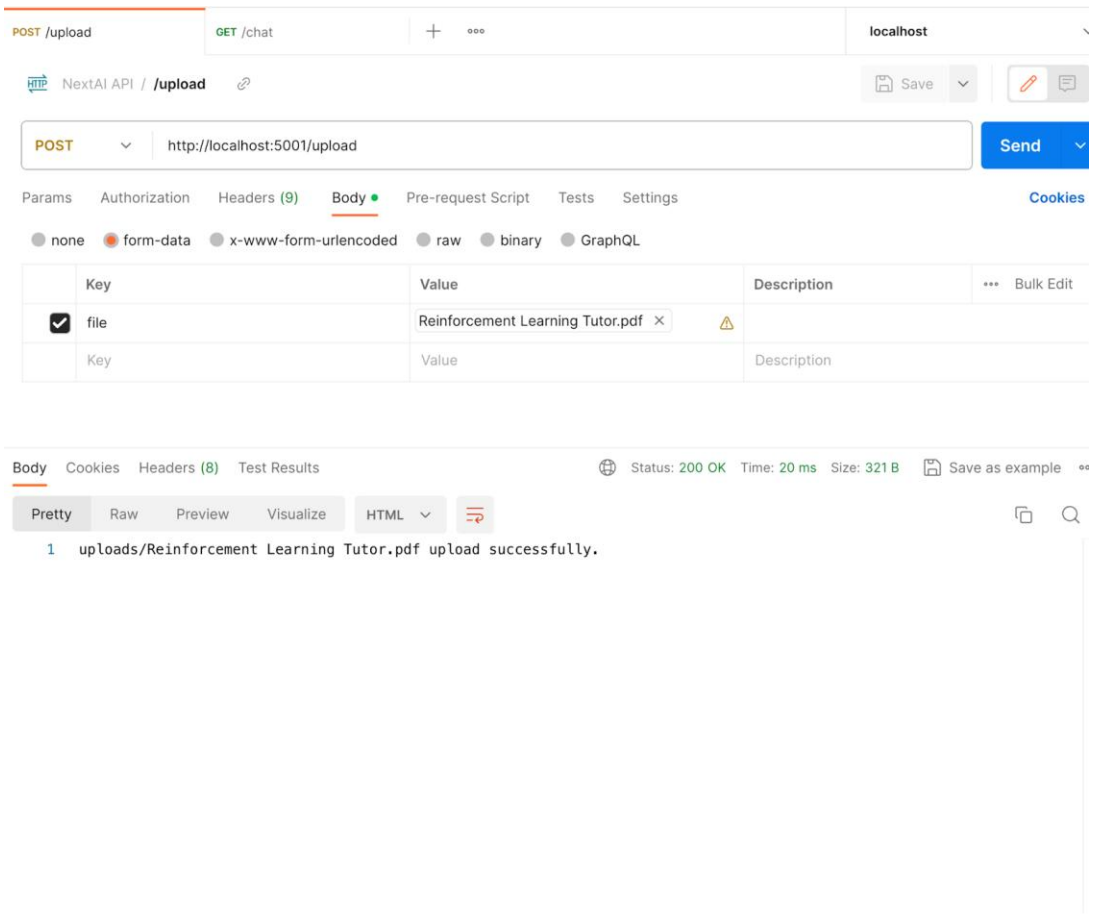
```
  // Create a simple chain using the prompt template
  const formattedPrompt = await prompt.format({
    context,
    question: query,
  });

  // Get response from the model
  const response = await model.invoke(formattedPrompt);

  return { text: response.content };
};

export default chat;
```

Create a file called **server.js** with the following content

```
import express from "express";
import cors from "cors";
import dotenv from "dotenv";
import multer from "multer"; // Import multer
import chat from "./chat.js";

dotenv.config();

const app = express();
app.use(cors());

// Configure multer
const storage = multer.diskStorage({
  destination: function (req, file, cb) {
    cb(null, "uploads/");
  },
  filename: function (req, file, cb) {
    cb(null, file.originalname);
  },
});
const upload = multer({ storage: storage });

const PORT = 5001;

let filePath;

app.post("/upload", upload.single("file"), (req, res) => {
  // Use multer to handle file upload
  filePath = req.file.path; // The path where the file is temporarily saved
  res.send(filePath + " upload successfully.");
});

app.get("/chat", async (req, res) => {
  const resp = await chat(filePath, req.query.question); // Use MCP-enhanced chat
  res.send({
    ragAnswer: resp.text,
    mcpAnswer: "N/A",
  });
});

app.listen(PORT, () => {
  console.log(`Server is running on port ${PORT}`);
});
```

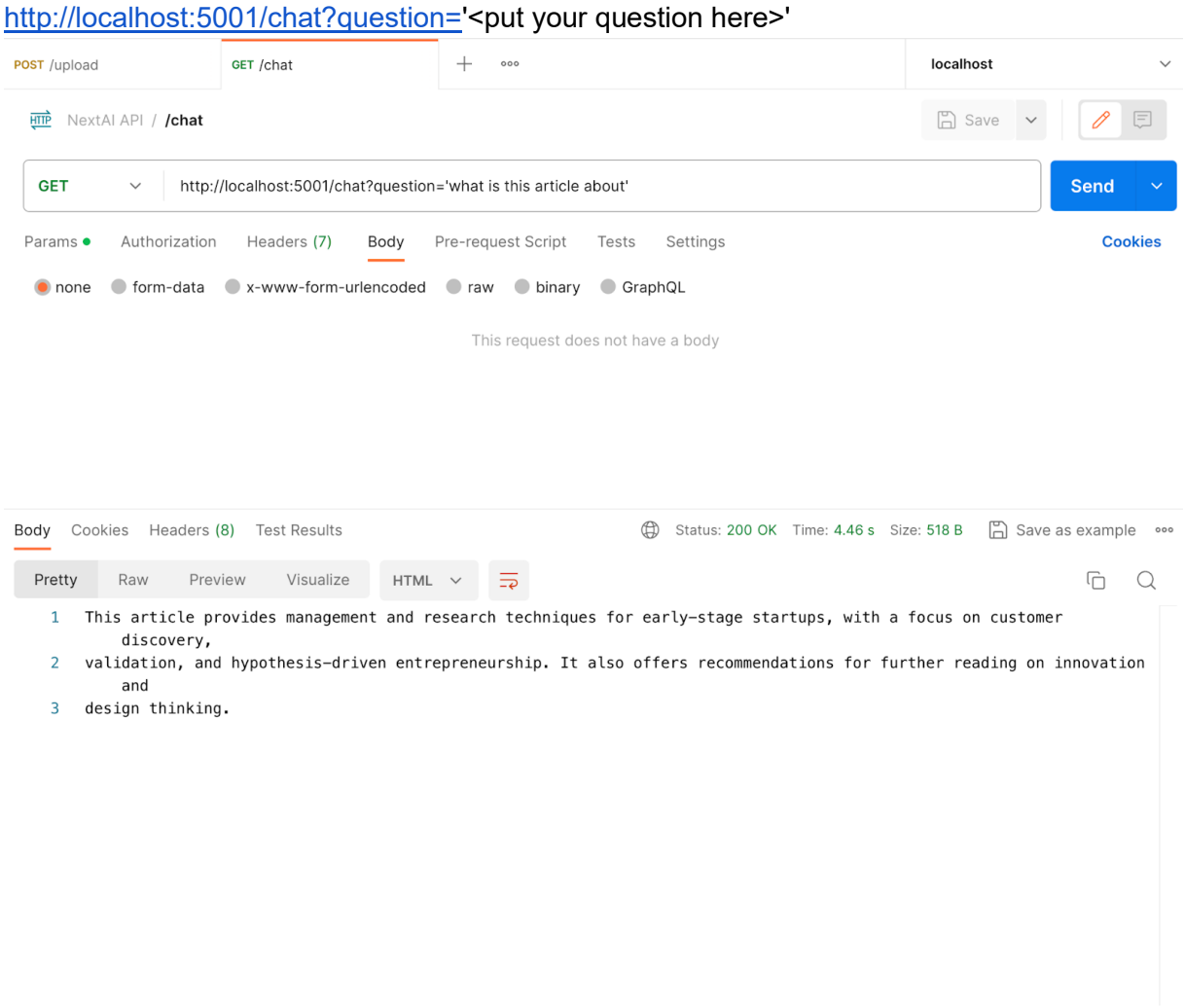
Now, let's run the server (`node server.js`) and use Postman to test our Express.js backend

Test the /upload endpoint
http://localhost:5001/upload

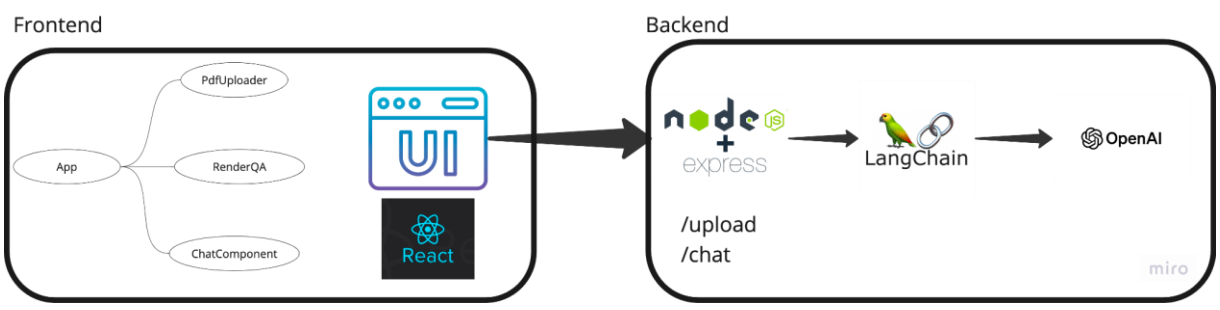


Test the /chat endpoint

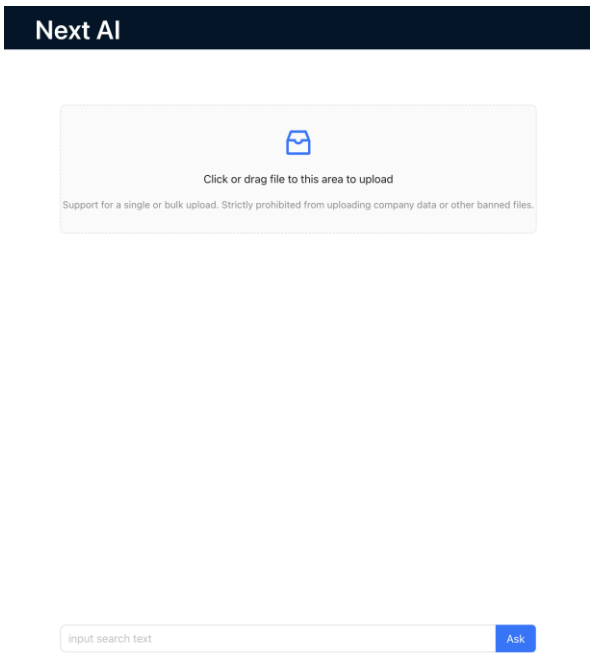
http://localhost:5001/chat?question='<put your question here>'



Build the React UI



By the end of this class, you will have the following working app.



Under root, update **package.json** with the followings

```
{
"name": "agentai",
"version": "0.1.0",
"private": true,
"dependencies": {
  "@testing-library/jest-dom": "^5.17.0",
  "@testing-library/react": "^13.4.0",
  "@testing-library/user-event": "^13.5.0",
  "antd": "^5.8.4",
  "axios": "^1.5.0",
  "concurrently": "^8.2.1",
  "react": "^18.2.0",
  "react-dom": "^18.2.0",
  "react-scripts": "5.0.1",
  "web-vitals": "^2.1.4"
},
"scripts": {
  "dev": "concurrently \"npm start\" \"npm run server\"",
  "server": "cd server && npm run start",
  "start": "react-scripts start",
  "build": "react-scripts build",
  "test": "react-scripts test",
  "eject": "react-scripts eject"
},
"eslintConfig": {
"extends": [
  "react-app",
  "react-app/jest"
]
},
"browserslist": {
  "production": [
    ">0.2%",
    "not dead",
    "not op_mini all"
],
"development": [
      "last 1 chrome version",
      "last 1 firefox version",
      "last 1 safari version"
    ]
  }
}
```

Next, let's install all dependencies by running `**npm install**`

Under the root folder, create a .env file with the following content.

```
.env
```

Under **src,** create a folder called `**components**`

src/components/ChatComponent.js

```
import React, { useState } from "react"; // Import useState
import axios from "axios";
import { Input } from "antd";

const { Search } = Input;

const DOMAIN = "http://localhost:5001";

const searchContainer = {
  display: "flex",
  justifyContent: "center",
};

const ChatComponent = (props) => {
  const { handleResp, isLoading, setIsLoading } = props;
  // Define a state variable to keep track of the search value
  const [searchValue, setSearchValue] = useState("");

  const onSearch = async (question) => {
    // Clear the search input
    setSearchValue("");
    setIsLoading(true);

    try {
      const response = await axios.get(`${DOMAIN}/chat`, {
        params: {
          question,
        },
      });
      handleResp(question, response.data);
    } catch (error) {
      console.error(`Error: ${error}`);
      handleResp(question, error);
    } finally {
      setIsLoading(false);
    }
  };

  const handleChange = (e) => {
    // Update searchValue state when the user types in the input box
    setSearchValue(e.target.value);
  };

  return (
    <div style={searchContainer}>
      <Search
        placeholder="input search text"
        enterButton="Ask"
        size="large"
        onSearch={onSearch}
        loading={isLoading}
        value={searchValue} // Control the value
        onChange={handleChange} // Update the value when changed
      />
    </div>
  );
};

export default ChatComponent;
```

src/components/PdfUploader.js

```
import React from "react";
import axios from "axios"; // Import axios for HTTP requests
import { InboxOutlined } from "@ant-design/icons";
import { message, Upload } from "antd";

const { Dragger } = Upload;

const DOMAIN = process.env.REACT_APP_DOMAIN;

const uploadToBackend = async (file) => {
  const formData = new FormData();
  formData.append("file", file);
  try {
```

```
    const response = await axios.post(`${DOMAIN}/upload`, formData, {
      headers: {
        "Content-Type": "multipart/form-data",
      },
    });
    return response;
  } catch (error) {
    console.error("Error uploading file: ", error);
    return null;
  }
};

const attributes = {
  name: "file",
  multiple: true,
  customRequest: async ({ file, onSuccess, onError }) => {
    const response = await uploadToBackend(file);
    if (response && response.status === 200) {
      // Handle success
      onSuccess(response.data);
    } else {
      // Handle error
      onError(new Error("Upload failed"));
    }
  },
  onChange(info) {
    const { status } = info.file;
    if (status !== "uploading") {
      console.log(info.file, info.fileList);
    }
    if (status === "done") {
      message.success(`${info.file.name} file uploaded successfully.`);
    } else if (status === "error") {
      message.error(`${info.file.name} file upload failed.`);
    }
  },
  onDrop(e) {
    console.log("Dropped files", e.dataTransfer.files);
  },
};

const PdfUploader = () => {
  return (
    <Dragger {...attributes}>
      <p className="ant-upload-drag-icon">
        <InboxOutlined />
      </p>
      <p className="ant-upload-text">
        Click or drag file to this area to upload
      </p>
      <p className="ant-upload-hint">
        Support for a single or bulk upload. Strictly prohibited from uploading
        company data or other banned files.
      </p>
    </Dragger>
  );
};

export default PdfUploader;
```

src/components/RenderQA.js

```
import React from "react";
import { Spin } from "antd";

const containerStyle = {
  display: "flex",
  justifyContent: "space-between",
  flexDirection: "column",
  marginBottom: "20px",
};

const userContainer = {
  textAlign: "right",
};

const agentContainer = {
  textAlign: "left",
};

const userStyle = {
  maxWidth: "50%",
  textAlign: "left",
  backgroundColor: "#1677FF",
  color: "white",
```

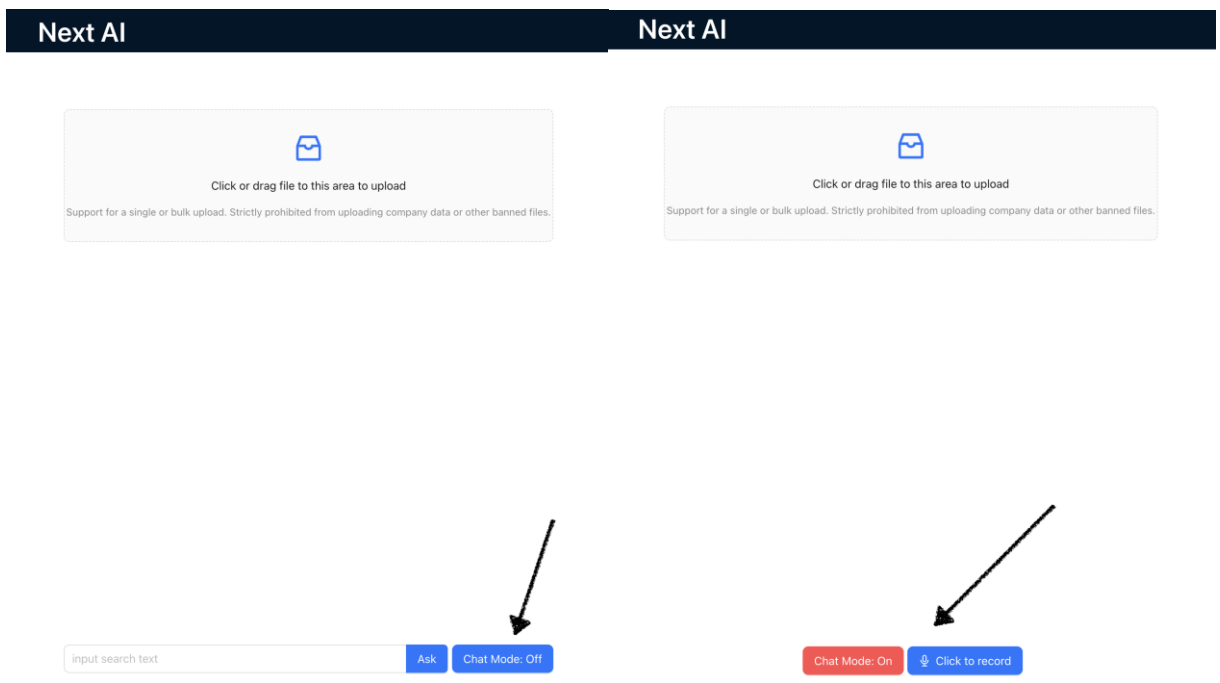
```
  display: "inline-block",
  borderRadius: "10px",
  padding: "10px",
  marginBottom: "10px",
};

const answerContainer = {
  marginBottom: "10px",
};

const answerLabel = {
  fontSize: "12px",
  fontWeight: "bold",
  color: "#666",
  marginBottom: "5px",
};

const ragAnswerStyle = {
  maxWidth: "50%",
  textAlign: "left",
  backgroundColor: "#E6F7FF",
  color: "black",
  display: "inline-block",
  borderRadius: "10px",
  padding: "10px",
  marginBottom: "5px",
  borderLeft: "4px solid #1890FF",
};

const mcpAnswerStyle = {
  maxWidth: "50%",
  textAlign: "left",
  backgroundColor: "#F6FFED",
  color: "black",
  display: "inline-block",
  borderRadius: "10px",
  padding: "10px",
  marginBottom: "5px",
  borderLeft: "4px solid #52C41A",
};

const RenderQA = (props) => {
  const { conversation, isLoading } = props;

  return (
    <>
      {conversation?.map((each, index) => {
        return (
          <div key={index} style={containerStyle}>
            <div style={userContainer}>
              <div style={userStyle}>{each.question}</div>
            </div>
            <div style={agentContainer}>
              <div>
                <div style={answerContainer}>
                  <div style={answerLabel}>RAG Answer (from document):</div>
                  <div style={ragAnswerStyle}>{each.answer.ragAnswer}</div>
                </div>
                <div style={answerContainer}>
                  <div style={answerLabel}>MCP Answer (with web search):</div>
                  <div style={mcpAnswerStyle}>{each.answer.mcpAnswer}</div>
                </div>
              </div>
            </div>
          </div>
        );
      })}
      {isLoading && <Spin size="large" style={{ margin: "10px" }} />}
    </>
  );
};

export default RenderQA;
```

Last, let's pull everything together by updating **App.js**

```
import React, { useState } from "react";
import PdfUploader from "./components/PdfUploader";
import ChatComponent from "./components/ChatComponent";
import RenderQA from "./components/RenderQA";
import { Layout, Typography } from "antd";

const chatComponentStyle = {
  position: "fixed",
```

```
  bottom: "0",
  width: "80%",
  left: "10%", // this will center it because it leaves 10% space on each side
  marginBottom: "20px",
};

const pdfUploaderStyle = {
  margin: "auto",
  paddingTop: "80px",
};

const renderQAStyle = {
  height: "50%", // adjust the height as you see fit
  overflowY: "auto",
};

const App = () => {
  const [conversation, setConversation] = useState([]);
  const [isLoading, setIsLoading] = useState(false);
  const { Header, Content } = Layout;
  const { Title } = Typography;

  const handleResp = (question, answer) => {
     setConversation((prev) => [...prev, { question, answer }]);
  };

  return (
    <>
      <Layout style={{ height: "100vh", backgroundColor: "white" }}>
        <Header
          style={{
            display: "flex",
            alignItems: "center",
          }}
        >
          <Title style={{ color: "white " }}>Agent AI</Title>
        </Header>
        <Content style={{ width: "80%", margin: "auto" }}>
          <div style={pdfUploaderStyle}>
            <PdfUploader />
          </div>

          <br />
          <br />
          <div style={renderQAStyle}>
            <RenderQA conversation={conversation} isLoading={isLoading} />
          </div>

          <br />
          <br />
        </Content>
        <div style={chatComponentStyle}>
          <ChatComponent
            handleResp={handleResp}
            isLoading={isLoading}
            setIsLoading={setIsLoading}
          />
        </div>
      </Layout>
    </>
  );
};

export default App;
```

Add the Voice Interface



Add the Conversational Interface

First, we need to install a few dependencies

npm i react-speech-recognition@3.10.0
npm i speak-tts@2.0.8

Next, we will update the src/components/ChatComponent.js

```
import React, { useState, useEffect } from "react"; // Import useState
import axios from "axios";
import { Button, Input } from "antd";
import { AudioOutlined } from "@ant-design/icons";
import SpeechRecognition, {
  useSpeechRecognition,
} from "react-speech-recognition";
import Speech from "speak-tts";

const { Search } = Input;

const DOMAIN = "http://localhost:5001";

const searchContainer = {
  display: "flex",
  justifyContent: "center",
};

const ChatComponent = (props) => {
  const { handleResp, isLoading, setIsLoading } = props;
  // Define a state variable to keep track of the search value
  const [searchValue, setSearchValue] = useState("");
  const [isChatModeOn, setIsChatModeOn] = useState(false);
  const [isRecording, setIsRecording] = useState(false);
  const [speech, setSpeech] = useState();

  // speech recognation
  const {
    transcript,
    listening,
    resetTranscript,
    browserSupportsSpeechRecognition,
    isMicrophoneAvailable,
  } = useSpeechRecognition();

  useEffect(() => {
    const initialized_speech = new Speech();
    initialized_speech
```

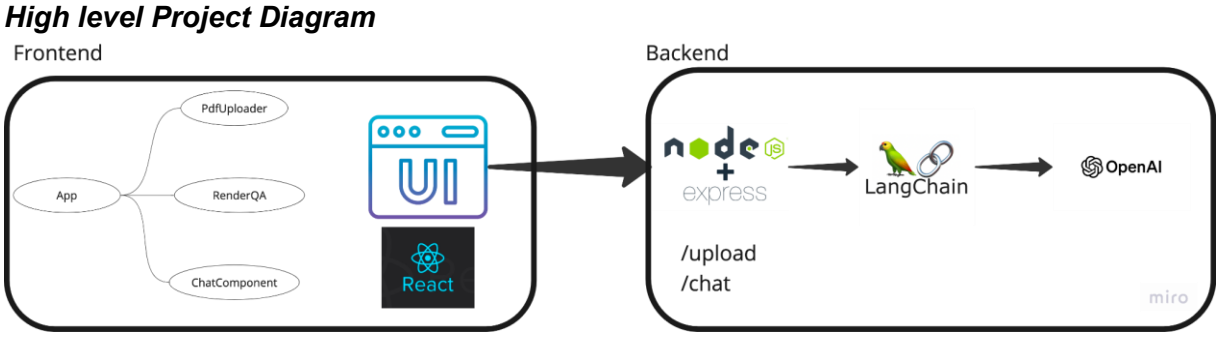
```
      .init({
        volume: 1,
        lang: "en-US",
        rate: 1,
        pitch: 1,
        voice: "Google US English",
        splitSentences: false,
      })
      .then((data) => {
        // The "data" object contains the list of available voices and the voice synthesis params
        console.log("Speech is ready, voices are available", data);
        setSpeech(initialized_speech);
      })
      .catch((e) => {
        console.error("An error occured while initializing : ", e);
      });
  }, []);

  useEffect(() => {
    if (!listening && Boolean(transcript)) {
      (async () => await onSearch(transcript))();
      setIsRecording(false);
    }
  }, [listening, transcript]);

  const talk = (what2say) => {
    speech
      .speak({
        text: what2say,
        queue: false, // current speech will be interrupted,
        listeners: {
          onstart: () => {
            console.log("Start utterance");
          },
          onend: () => {
            console.log("End utterance");
          },
          onresume: () => {
            console.log("Resume utterance");
          },
          onboundary: (event) => {
            console.log(
              event.name +
                " boundary reached after " +
                event.elapsedTime +
                " milliseconds."
            );
          },
        },
      })
      .then(() => {
        // if everyting went well, start listening again
        console.log("Success !");
        userStartConvo();
      })
      .catch((e) => {
        console.error("An error occurred :", e);
      });
  };

  const userStartConvo = () => {
    SpeechRecognition.startListening();
    setIsRecording(true);
    resetTranscript();
  };

  const chatModeClickHandler = () => {
    setIsChatModeOn(!isChatModeOn);
    setIsRecording(false);
```

```
    SpeechRecognition.stopListening();

    resetTranscript();
  };

  const recordingClickHandler = () => {
    if (isRecording) {
      setIsRecording(false);
      SpeechRecognition.stopListening();

      resetTranscript();
    } else {
      setIsRecording(true);
      SpeechRecognition.startListening();
    }
  };

  const onSearch = async (question) => {
    // Clear the search input
    setSearchValue("");
    setIsLoading(true);

    try {
      const response = await axios.get(`${DOMAIN}/chat`, {
        params: {
          question,
        },
      });
      handleResp(question, response.data);
      if (isChatModeOn) {
        talk(response.data?.ragAnswer);
      }
    } catch (error) {
      console.error(`Error: ${error}`);
      handleResp(question, error);
    } finally {
      setIsLoading(false);
    }
  };

  const handleChange = (e) => {
    // Update searchValue state when the user types in the input box
    setSearchValue(e.target.value);
  };

  return (
    <div style={searchContainer}>
      {!isChatModeOn && (
        <Search
          placeholder="input search text"
          enterButton="Ask"
          size="large"
          onSearch={onSearch}
          loading={isLoading}
          value={searchValue} // Control the value
          onChange={handleChange} // Update the value when changed
        />
      )}
      <Button
        type="primary"
        size="large"
        danger={isChatModeOn}
        onClick={chatModeClickHandler}
        style={{ marginLeft: "5px" }}
      >
        Chat Mode: {isChatModeOn ? "On" : "Off"}
      </Button>
      {isChatModeOn && (
        <Button
```

```
          type="primary"
          icon={<AudioOutlined />}
          size="large"
          danger={isRecording}
          onClick={recordingClickHandler}
          style={{ marginLeft: "5px" }}
        >
          {isRecording ? "Recording..." : "Click to record"}
        </Button>
      )}
    </div>
  );
};

export default ChatComponent;
```

That's it for the Agent AI Project. Now, have fun uploading documents and start chatting with the PDF!

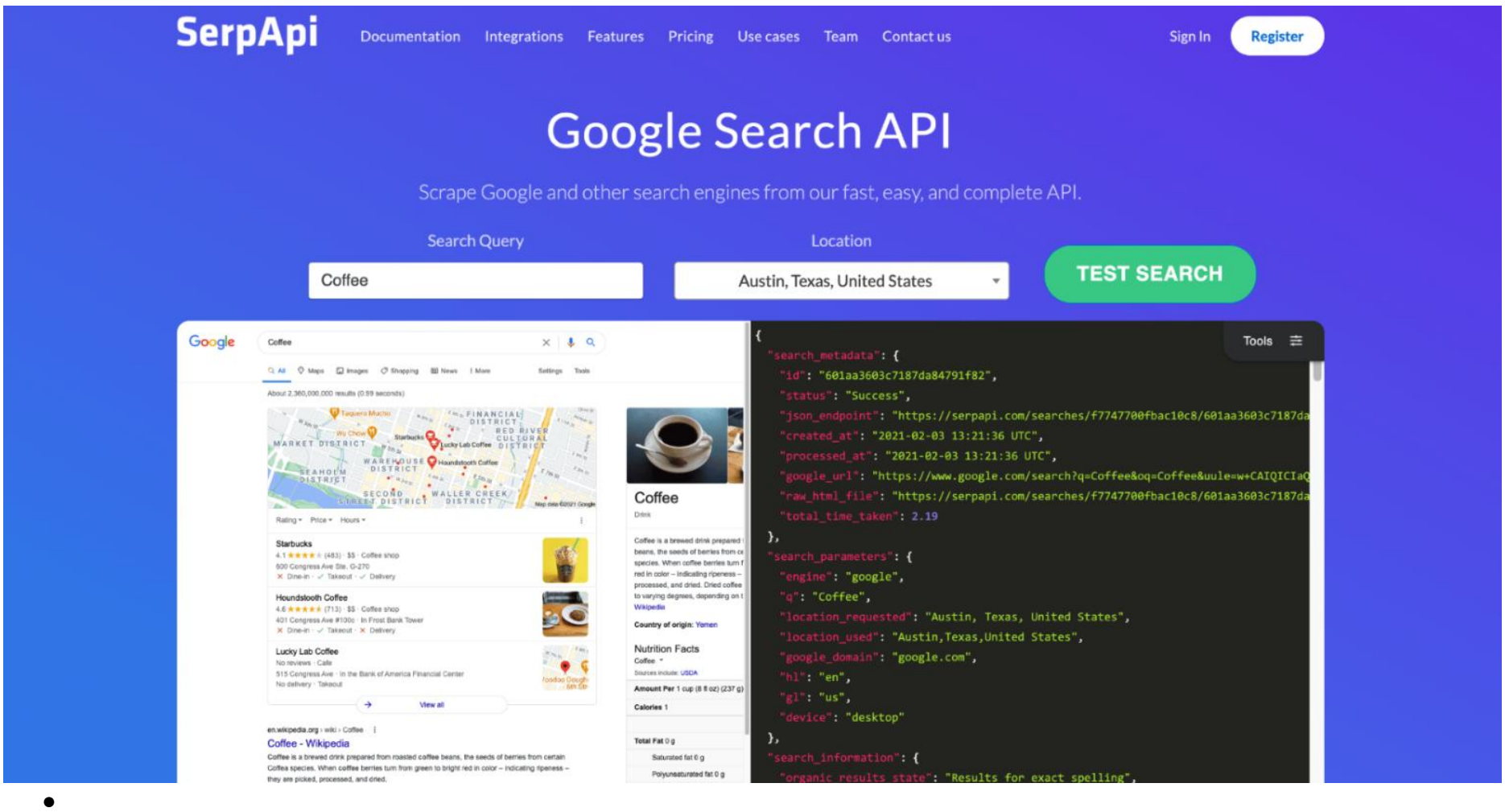
High level Project Diagram



Non-Functional Requirements

- (optional) deploy the application to AWS/GCP
  - e.g. AWS Amplify (frontend), AWS App Runner (backend)
- consider non-functional requirements (NFR)
  - Performance
  - Scalability
  - Reliability
  - Availability
  - Maintainability
  - Security
  - Usability
  - Legal
    - Accessibility (aria-label)

Next, let's get the serpAPI



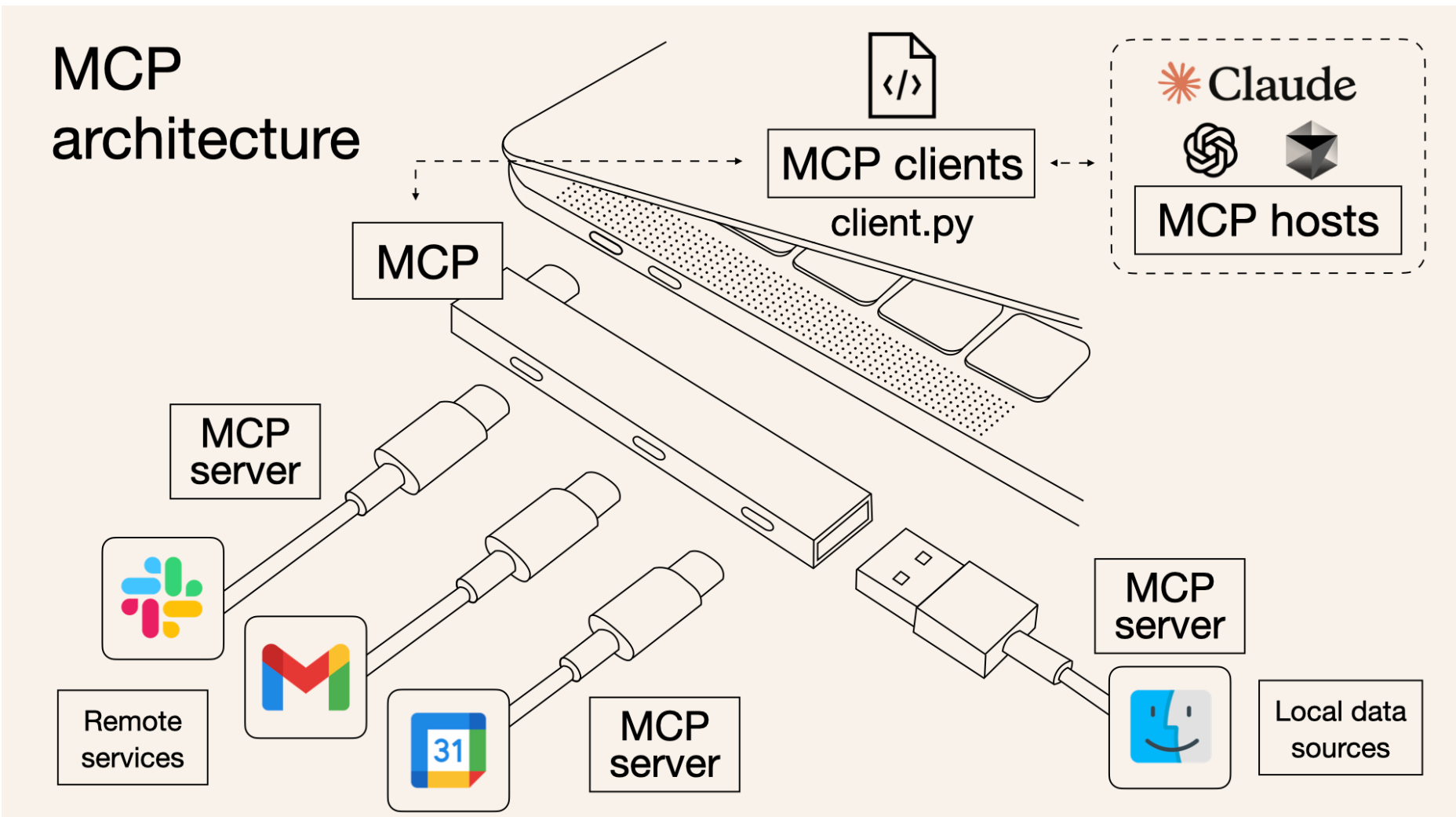
•

Open **server/.env** file and add your key there

SERPAPI_KEY=<your_key>

Model Context Protocol (MCP) Integration

What is MCP



Our architecture

HTTP Request → server.js → MCP Client (chat-mcp.js) → MCP Server (child process, Provides the search_web tool)

Let's create a file in server/mcp-server.js (MCP SERVER)

```
import { McpServer } from "@modelcontextprotocol/sdk/server/mcp.js";
```

```
import { StdioServerTransport } from "@modelcontextprotocol/sdk/server/stdio.js";
import { z } from "zod";
import { getJson } from "serpapi";

const SERPAPI_KEY = process.env.SERPAPI_KEY;

// Create MCP server instance
const server = new McpServer({
  name: "serpapi-search",
  version: "1.0.0",
});

// Register the search tool
server.registerTool(
  "search_web",
  {
    description:
      "Search the web using SerpAPI. Returns search results including organic results, snippets, and related
information.",
    inputSchema: {
      query: z.string().describe("The search query to execute"),
      num: z
        .number()
        .optional()
        .describe("Number of results to return (default: 10)"),
    },
  },
  async ({ query, num = 10 }) => {
    try {
      const results = await getJson({
        engine: "google",
        q: query,
        num: num,
        api_key: SERPAPI_KEY,
      });

      // Return all results as plain text
      const fullResults = JSON.stringify(results);

      return {
        content: [
          {
            type: "text",
            text: fullResults,
          },
        ],
      };
    } catch (error) {
      return {
        content: [
          {
            type: "text",
            text: `Error performing web search: ${error.message}`,
          },
        ],
      };
    }
  }
);

// Main function to run the server
async function main() {
  const transport = new StdioServerTransport();
  await server.connect(transport);
  console.error("SerpAPI MCP Server running on stdio");
}

// Always run the server when this file is executed
// This is needed when spawned as a child process
```

```
main().catch((error) => {
  console.error("Fatal error in MCP server:", error);
  process.exit(1);
});

export default server;
```

Next, let's create server/chat-mcp.js **(MCP CLIENT)**

```
import { Client } from "@modelcontextprotocol/sdk/client/index.js";
import { StdioClientTransport } from "@modelcontextprotocol/sdk/client/stdio.js";
import { fileURLToPath } from "url";
import { dirname, join } from "path";
import { ChatOpenAI } from "@langchain/openai";
import { PromptTemplate } from "@langchain/core/prompts";

const __filename = fileURLToPath(import.meta.url);
const __dirname = dirname(__filename);

// Reusable MCP client instance
let client = null;
let transport = null;
let isConnecting = false;
let connectionPromise = null;

// Initialize the MCP client connection (lazy initialization)
const ensureConnected = async () => {
  // If already connected, return
  if (client && transport) {
    return;
  }

  // If connection is in progress, wait for it
  if (isConnecting && connectionPromise) {
    await connectionPromise;
    return;
  }

  // Start new connection
  isConnecting = true;
  connectionPromise = (async () => {
    try {
      // Create MCP client
      client = new Client({
        name: "chat-client",
        version: "1.0.0",
      });

      // Get the path to the MCP server
      const serverPath = join(__dirname, "mcp-server.js");

      // Create transport to connect to the MCP server
      transport = new StdioClientTransport({
        command: "node",
        args: [serverPath],
        env: {
          ...process.env, // Inherit all environment variables from parent process
        },
      });

      // Connect to the MCP server
      await client.connect(transport);
    } finally {
      isConnecting = false;
      connectionPromise = null;
    }
  })();
```

```
  await connectionPromise;
};

const chatMCP = async (query) => {
  const apiKey = process.env.OPENAI_API_KEY;

  try {
    // Ensure client is connected (reuse existing connection)
    await ensureConnected();

    // Always perform web search
    const toolResult = await client.callTool({
      name: "search_web",
      arguments: {
        query: query,
        num: 5,
      },
    });

    let searchResults = "";

    if (toolResult.content && toolResult.content.length > 0) {
      searchResults = toolResult.content[0].text;
    }

    // Use the LLM to answer the question based on search results
    const model = new ChatOpenAI({
      model: "gpt-5",
      apiKey,
    });

    const answerTemplate = `Summarize the search result.

Search Results:
{searchResults}

Helpful Answer:`;

    const prompt = PromptTemplate.fromTemplate(answerTemplate);
    const formattedPrompt = await prompt.format({
      searchResults: searchResults || "No search results available",
    });

    const response = await model.invoke(formattedPrompt);
    const finalAnswer = response.content;

    return { text: finalAnswer };
  } catch (error) {
    // If connection error, reset client to allow reconnection on next request
    if (client) {
      try {
        await client.close();
      } catch (e) {
        // Ignore cleanup errors
      }
      client = null;
      transport = null;
    }

    throw error;
  }
};

export default chatMCP;
```

Now, last step, let's modify server.js

```
import express from "express";
import cors from "cors";
import dotenv from "dotenv";
import multer from "multer";
import chat from "./chat.js";
import chatMCP from "./chat-mcp.js";

dotenv.config();

const app = express();
app.use(cors());

const storage = multer.diskStorage({
  destination: (req, file, cb) => {
    cb(null, "uploads/");
  },
  filename: (req, file, cb) => {
    cb(null, file.originalname);
  },
});

const upload = multer({
  storage,
});

const PORT = 5001;

let filePath;

app.post("/upload", upload.single("file"), (req, res) => {
  filePath = req.file.path;
  res.send(filePath + "uploaded successfully");
});

app.get("/chat", async (req, res) => {
  const ragResp = await chat(filePath, req.query.question);
  const mcpResp = await chatMCP(req.query.question);

  res.send({
    ragAnswer: ragResp.text,
    mcpAnswer: mcpResp.text,
  });
});

app.listen(PORT, () => {
  console.log("server is running on port " + PORT);
});
```